



Unsupervised Concept Discovery and Evaluation

for Medical Vision-Language Models

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Presenter: Emmanuel.

Good morning everyone. Today we present our work on Unsupervised Concept Discovery and Evaluation for Medical Vision-Language Models, focusing on extracting interpretable features from chest X-ray embeddings.



Introduction & Abstract

Unsupervised Concept Discovery in VLMs

- **Objective:** Investigate unsupervised concept discovery in medical Vision-Language Models (VLMs) without diagnostic supervision[cite: 3].
- **Methodology:** A Sparse Autoencoder (SAE) decomposes frozen BiomedCLIP embeddings into 1,024 latent features[cite: 3].
- **Vocabulary Generation:** Features are named using 998 phrases associated with 100 clinical concepts, extracted without relying on downstream disease labels[cite: 3].
- **Semantic Evaluation:** Extracted concepts are assessed against Open-I radiology reports using a frozen MedGemma evaluator[cite: 3].
- **Key Findings:**
 - SAE achieves near-lossless reconstruction with high sparsity, but textual alignment remains low on average (7.21% Aligned)[cite: 3].
 - Alignment is highly uneven across clinical subdomains[cite: 3].
 - Evaluator reliability is a critical bottleneck: MedGemma agreed with human consensus only 56% of the time[cite: 3].

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Unsupervised Concept Discovery and Evaluation

Introduction & Abstract

Good morning everyone. Today we are presenting our analysis on unsupervised concept discovery in medical Vision-Language Models[cite: 3]. Our main objective is to understand the clinical meaning of latent dimensions in models like BiomedCLIP without relying on manual diagnostic labels[cite: 3]. To do this, we trained a Sparse Autoencoder to decompose image embeddings into 1,024 features, which we then named using a custom vocabulary of 998 medical phrases[cite: 3]. We evaluated these concepts against real radiology reports using MedGemma as an LLM judge[cite: 3]. Our core finding is that while we can achieve excellent and sparse visual reconstruction, this does not automatically translate to high clinical textual agreement[cite: 3]. Furthermore, the choice of the LLM judge heavily impacts the evaluation results[cite: 3].

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Related Work

From Predefined Concepts to Unsupervised Discovery

- **Concept Representation:**

- Methods like TCAV and Concept Bottleneck Models provide semantic interfaces but strictly require user-defined, annotated concepts[cite: 3].
- Sparse dictionary learning decomposes polysemantic activations into monosemantic features without predefined labels[cite: 3].

- **VLMs and Medical Grounding:**

- Recent works dissect neuron activations to build interpretable medical explanations directly from opaque networks[cite: 3].
- **MedConcept:** Combines sparse decomposition, ontology-derived vocabularies, and LLM judgments[cite: 3]. We adapt this pipeline from 3D abdominal CTs to 2D chest radiographs using BiomedCLIP[cite: 3].

- **Evaluating Explanations:**

- Reconstruction fidelity, human inspection, and report entailment are complementary evaluation metrics[cite: 3].
- LLM-as-a-judge offers scalability but introduces vulnerabilities like position bias[cite: 3].

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Related Work

Moving to related work, traditional interpretability approaches like Concept Bottleneck Models rely heavily on predefined, manual annotations[cite: 3]. To bypass this, recent advancements in Sparse Dictionary Learning allow us to extract concepts in a completely unsupervised way[cite: 3]. Our specific methodology builds upon the MedConcept framework[cite: 3]. However, we transition the focus from 3D CT scans to 2D chest X-rays, replacing the backbone with BiomedCLIP and building a new dedicated vocabulary[cite: 3]. Finally, evaluating these discovered explanations remains an open challenge[cite: 3]. While using an LLM as a judge scales well, literature shows it introduces inherent biases that require independent human validation[cite: 3].

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- **VLMs and Medical Grounding:**
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Research Gaps

Limitations in Evaluating Interpretability

- **Reconstruction vs. Semantic Grounding:** High SAE sparsity and reconstruction fidelity do not guarantee that the nearest text label correctly describes the feature[cite: 3].
- **Dependence on External Resources:** The discovery pipeline relies heavily on external tools (ontology vocabularies, generative phrasing, LLM judges)[cite: 3]. Biases in these components can be mistaken for properties of the learned representation[cite: 3].
- **Selective Evidence in Reports:** Radiology reports provide partial evidence[cite: 3]. Unmentioned concepts are not necessarily absent, causing external judges to confuse omission with contradiction[cite: 3].
- **Independent Labeling:** Assigning labels independently omits causal and anatomical relationships between findings[cite: 3]. We explore semantic clustering to identify useful macro-groups[cite: 3].

Unsupervised Concept Discovery and Evaluation

Research Gaps

We identified four major research gaps that guide our analysis. First, mathematical reconstruction fidelity does not ensure semantic correctness; the geometry of the embedding space heavily constrains the naming process[cite: 3]. Second, the entire pipeline is dependent on external resources like UMLS ontologies and generative models[cite: 3]. Errors generated by these tools can easily be misattributed to the SAE itself[cite: 3]. Third, radiology reports are selective by nature[cite: 3]. Radiologists often omit normal anatomy, which an automated judge might wrongly penalize as a contradiction[cite: 3]. Lastly, treating concepts in isolation ignores their real-world anatomical relationships, a gap we attempt to address later through hierarchical clustering[cite: 3].

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Methodology

Datasets and Feature Extraction

- **NIH Chest X-ray 10k:** 7,500 images used exclusively to train the Sparse Autoencoder (SAE), and 2,500 for validation[cite: 3]. No text or labels were used[cite: 3].
- **Open-I:** 3,666 image-report pairs used for semantic evaluation[cite: 3]. Reports concatenate Findings and Impression sections[cite: 3].
- **Feature Extraction:** BiomedCLIP's frozen ViT-B/16 image encoder produced the visual embeddings for both datasets[cite: 3].

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Unsupervised Concept Discovery and Evaluation

Methodology

Let's discuss our methodology, starting with the datasets[cite: 3]. We kept discovery and evaluation strictly separate. We trained the Sparse Autoencoder on 7,500 images from the NIH dataset without any text or labels[cite: 3]. For evaluation, we used the Open-I dataset, which provides paired images and radiology reports[cite: 3]. All visual embeddings were extracted using BiomedCLIP's frozen ViT-B/16 encoder[cite: 3].

Methodology

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Methodology

Concept Dictionary Construction

- **Concept Selection:** 100 clinical parent concepts extracted and filtered from Open-I Problems metadata[cite: 3].
- **Expansion:** Concepts were mapped via UMLS to SNOMED CT and MeSH to retrieve synonyms and Unique Identifiers (CUIs)[cite: 3].
- **Phrase Generation:** Gemini 3.1 Flash-Lite generated 5 descriptive phrases per concept[cite: 3].
- **Encoding:** Phrases were encoded with BiomedCLIP's PubMedBERT into a 998×512 matrix[cite: 3].

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Unsupervised Concept Discovery and Evaluation

Methodology

To name the features learned by the SAE, we built a dictionary of medical concepts[cite: 3]. We started with 100 parent concepts from Open-I metadata and linked them to standardized vocabularies like SNOMED CT using UMLS[cite: 3]. To get radiology-style descriptions, we used Gemini 3.1 Flash-Lite to generate 5 short phrases per concept[cite: 3]. These phrases were then encoded with PubMedBERT to map them into the same latent space as the images[cite: 3].

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Methodology

Sparse Autoencoder and Concept Assignment

- **SAE Architecture:** Input dimension $d = 512$, hidden dimension $m = 1024$ (overcomplete factor of 2)[cite: 3].
- **Training:** Optimization includes an L_1 penalty to encourage sparse activations[cite: 3].
- **Naming:** Each learned feature direction (decoder column) is assigned a phrase based on the highest cosine similarity[cite: 3].
- **Inference:** For a new image, active neurons contributing to $> 90\%$ of the total positive activation are selected to form the predicted concept set[cite: 3].

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Methodology

Our Sparse Autoencoder maps the 512-dimensional embeddings into a 1,024-dimensional hidden space[cite: 3]. By applying an L1 penalty, we force the model to use only a few active units per image[cite: 3]. We assigned names to these 1,024 features by finding the closest phrase in our dictionary using cosine similarity[cite: 3]. During inference, we select the top features that account for 90 percent of the positive activation for a given image[cite: 3].

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Methodology

Semantic Evaluation and Validation

- **Evaluator:** Frozen MedGemma 1.5 4B[cite: 3].
- **Verdicts:**
 - **Aligned:** Report explicitly supports the concept[cite: 3].
 - **Unaligned:** Report explicitly negates/contradicts the concept[cite: 3].
 - **Uncertain:** Concept is unmentioned or ambiguous[cite: 3].
- **Bias Mitigation:** 3 forward passes with permuted label-to-letter mappings to reduce position bias[cite: 3].

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Methodology

To evaluate if the extracted concepts match the radiology reports, we used MedGemma 1.5 4B as a judge[cite: 3]. The model classifies each concept-report pair as Aligned, Unaligned, or Uncertain[cite: 3]. To prevent the LLM from simply favoring the first option, we rotated the answer choices across three separate passes and averaged the probabilities[cite: 3].

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Semantic Evaluation and Validation

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Experiments

Sparse Reconstruction and Grounding

- **Reconstruction Fidelity:** Mean Squared Error (MSE) of 3.32×10^{-5} and mean cosine similarity of 0.9916 on the validation set[cite: 3].
- **Sparsity:** Average of 44.0 active units out of 1024 ($\sim 4.3\%$ L_0 sparsity) per image[cite: 3].
- **Takeaway:** The SAE achieves near-lossless visual reconstruction under a strict sparse activation regime[cite: 3]. However, fidelity does not guarantee semantic monosemanticity[cite: 3].

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Experiments

Moving to our results, the SAE training was highly successful in terms of reconstruction[cite: 3]. We achieved a very low Mean Squared Error and a cosine similarity above 0.99 on the validation set[cite: 3]. It did this while maintaining high sparsity, using an average of only 44 active units out of 1024 per image[cite: 3]. However, as we will see, this mathematical fidelity does not automatically mean the features are perfectly interpretable clinically[cite: 3].

Experiments

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Experiments

Report Alignment and Cohort Analysis

- **Cohort Performance** ($N = 500$):
 - **Uncertain:** 70.30%[\[cite: 3\]](#)
 - **Unaligned:** 22.49%[\[cite: 3\]](#)
 - **Aligned:** 7.21%[\[cite: 3\]](#)
- **Omission Bias:** The dominance of Uncertain verdicts is due to radiologists omitting normal anatomy and unmentioned concepts[\[cite: 3\]](#).
- **Unaligned Rate:** Captures both false-positive concept predictions and explicit clinical negations[\[cite: 3\]](#).

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When we tested the extracted concepts against actual reports on a 500-image cohort, the alignment was surprisingly low, averaging about 7.2 percent[\[cite: 3\]](#). Over 70 percent of concepts were marked as Uncertain[\[cite: 3\]](#). This is heavily driven by reporting omission bias: radiologists document acute pathology but skip normal anatomy, which the SAE still extracts[\[cite: 3\]](#). The 22.5 percent Unaligned rate includes both model false-positives and explicit negations in the text[\[cite: 3\]](#).



Experiments

Hierarchical Concept Clustering

- **Approach:** Agglomerative hierarchical clustering on the 100 parent concept embeddings using cosine distance[cite: 3].
- **Localized Alignment:**
 - Focal, specific clusters score highly: *Costophrenic Angle / Sulcus* achieves **49.23%** Aligned[cite: 3].
 - Broad groups score poorly: *Mixed Cardiopulmonary* yields only 8.62% Aligned[cite: 3].
- **Conclusion:** Clinical agreement is highly localized and unevenly distributed across subdomains[cite: 3].

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Experiments

To see if alignment varied by topic, we clustered the 100 parent concepts[cite: 3]. We found that clinical agreement is highly localized[cite: 3]. For example, focal anatomical findings like the "Costophrenic Angle" achieved over 49 percent alignment[cite: 3]. Conversely, broad or mixed clusters performed very poorly[cite: 3]. This shows that the SAE learns certain specific features much better than general ones[cite: 3].

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Experiments

Evaluator Audit and Qualitative Case Studies

- **Human Consensus:** 3 human experts audited 25 concept-report pairs[cite: 3].
- **Model Agreement vs. Humans:**
 - **MedGemma 1.5 4B:** 56.0% agreement[cite: 3].
 - **Commercial LLMs** (GPT-5.6, Claude 5, Gemini 3.1 Pro): 68.0% — 76.0% agreement[cite: 3].
- **Failure Modes:** MedGemma suffered from negation parsing errors (e.g., misclassifying "no pleural effusion" as an Aligned "large effusion")[cite: 3].

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Experiments

Finally, we audited our judge model[cite: 3]. Three experts manually evaluated 25 concept-report pairs[cite: 3]. We discovered that MedGemma agreed with human consensus only 56 percent of the time, while commercial models like Gemini 3.1 Pro reached up to 76 percent[cite: 3]. A major failure mode for MedGemma was negation: it would read "no pleural effusion" and incorrectly mark the concept "large effusion" as Aligned[cite: 3]. This proves that automated judges cannot yet replace expert review[cite: 3].

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Thank you for listening!
Any questions?